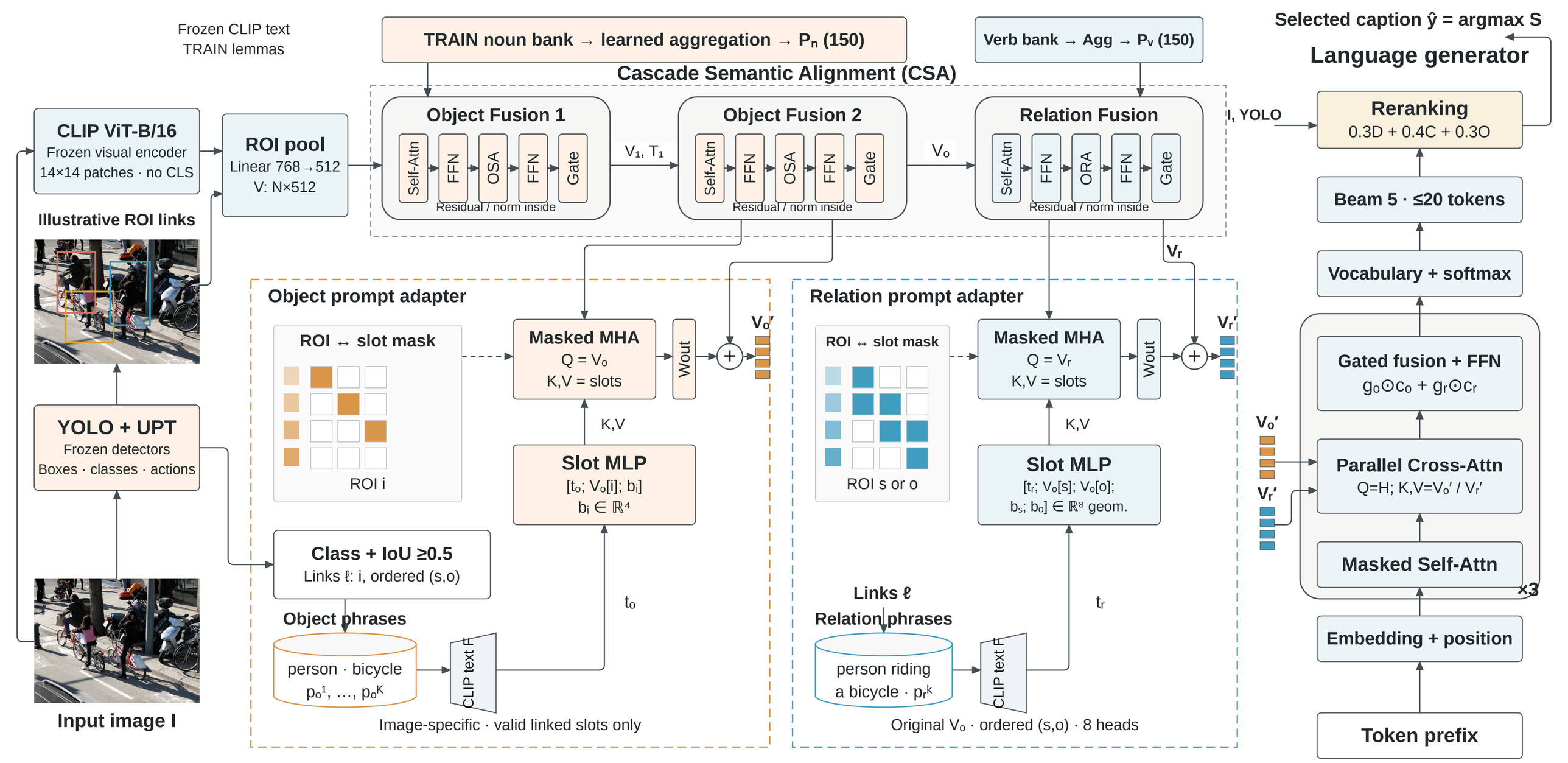

Frozen CLIP text
TRAIN lemmas
TRAIN noun bank → learned aggregation → Pn (150)
Verb bank → Agg → Pv (150)
Selected caption ŷ = argmax S
Language generator
Cascade Semantic Alignment (CSA)
CLIP ViT-B/16
Frozen visual encoder
14×14 patches · no CLS
ROI pool
Linear 768→512
V: N×512
Object Fusion 1
Self-Attn
FFN
OSA
FFN
Gate
Residual / norm inside
V1, T1
Object Fusion 2
Self-Attn
FFN
OSA
FFN
Gate
Residual / norm inside
Vo
Relation Fusion
Self-Attn
FFN
ORA
FFN
Gate
Residual / norm inside
I, YOLO
Reranking
0.3D + 0.4C + 0.3O
Beam 5 · ≤20 tokens
Vocabulary + softmax
Illustrative ROI links
Vr
Object prompt adapter
ROI ↔ slot mask
ROI i
Masked MHA
Q = Vo
K,V = slots
Wout
Vo′
K,V
Slot MLP
[to; Vo[i]; bi]
bi ∈ ℝ4
Class + IoU ≥0.5
Links ℓ: i, ordered (s,o)
Object phrases
person · bicycle
po1, …, poK
CLIP text F
to
Relation prompt adapter
ROI ↔ slot mask
ROI s or o
Masked MHA
Q = Vr
K,V = slots
Wout
Vr′
K,V
Slot MLP
[tr; Vo[s]; Vo[o];
bs; bo] ∈ ℝ8 geom.
Links ℓ
Relation phrases
person riding
a bicycle · prk
CLIP text F
tr
Image-specific · valid linked slots only
Original Vo · ordered (s,o) · 8 heads
Vo′
Vr′
Gated fusion + FFN
go⊙co + gr⊙cr
Parallel Cross-Attn
Q=H; K,V=Vo′ / Vr′
Masked Self-Attn
×3
Embedding + position
Token prefix
YOLO + UPT
Frozen detectors
Boxes · classes · actions
Input image I
TRAIN only: XE → XE + 0.05 SCST(CIDEr) | Wout zero-init; no valid slot ⇒ ΔV = 0
OSA / ORA: prototype alignment (τ=0.1) + cross-attention · F: frozen · Norm/residual details collapsed